

❏ 1. Live Forensics (Active State Analysis)

❏ Definition:

Live forensics is the process of **collecting volatile data** from a computer system that is **powered on and running**, before it is shut down or altered.

❏ Why Live Forensics?

In a powered-on system:

- Critical evidence exists in **volatile memory (RAM)**
 - Encryption keys may be **active in memory**
 - **Running processes, open network connections, and logs** may vanish if the system is shut down
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☒ Volatile Data Includes:

- RAM contents
- Running processes (ps, tasklist)
- Network connections (netstat, lsof)
- Loaded DLLs or kernel modules
- Clipboard contents
- Open files
- System uptime and logged-in users
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Cache, ARP table, routing table

☒ Tools Used in Live Forensics:

Platform	Tools
Windows	Sysinternals Suite, Volatility, DumpIt, FTK Imager (RAM capture), WinPMEM
Linux	dd, lsof, netstat, top, ps, LiME (Linux Memory Extractor), Volatility
Mac	OSX Collector, fs_usage, lsof, instruments

☒ Considerations:

- Always document every command and step taken

- Ensure **chain of custody** and **hash** all collected data
- Use **write blockers** for storage collection
- Avoid running **unknown binaries** on suspect machine
- Perform **memory acquisition first** – it changes the most

☒ 2. Linux Forensics (Artefact Collection & Analysis)

☒ Overview:

Linux forensics involves collecting and analyzing artefacts from a **Linux-based system** — particularly in server environments, where Linux dominates.

☒ Key Artefacts in Linux:

Artefact Type	Command/Location
Logs	/var/log/, /var/log/auth.log, /var/log/syslog, journalctl
Running Processes	ps aux, top, /proc
Scheduled Jobs	crontab -l, /etc/cron.*
User Activity	.bash_history, ~/.ssh/, w, last
Network Activity	netstat, ss, iptables -L
Mounted Devices	mount, lsblk, df -h
Open Files	lsof, /proc/<pid>/fd/
Memory Analysis	LiME, dd if=/dev/mem, Volatility
Installed Software	dpkg -l, rpm -qa
System Time	date, uptime

☒ Tools:

- **Autopsy and The Sleuth Kit** – For offline disk analysis
 - **Chkrootkit, rkhunter** – Rootkit detection
 - **Auditd** – Audit logs for system activity
 - **Strace** – Track system calls
 - **Volatility + LiME** – Memory acquisition and analysis
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☒ **Special Considerations:**

- Most Linux systems use **journaling filesystems** (ext4, XFS) – deleted files can often be recovered
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Properly manage **permissions** and **mount options** (e.g., noexec, nosuid) during analysis

- Preserve **metadata** and ensure **read-only mounts** for evidence drives

☒ 3. Introduction to Mobile Forensics

☒ What is Mobile Forensics?

Mobile forensics is the science of **recovering digital evidence** from **mobile devices** like smartphones and tablets.

☒ Scope Includes:

- Call logs
- SMS/MMS

- Contacts
- Images and videos
- App data (WhatsApp, Telegram, Instagram, etc.)
- Location data (GPS, cell tower logs)
- Web browsing history
- Deleted data and system logs

⌘ Challenges:

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Rapid evolution of mobile OS (Android/iOS)

- Strong encryption (Full Disk Encryption, Secure Enclave)
- Locked bootloaders
- App sandboxing
- Cloud backups and syncing
- Multiple file systems: YAFFS2, ext4, APFS, etc.

🔍 Tools Used in Mobile Forensics:

Tool	Platform	Description
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Cellebrite UFED	Android/iOS	Industry-standard for mobile data extraction
Magnet AXIOM	Android/iOS	Deep mobile + cloud + app analysis
Oxygen Forensics Detective	Android/iOS	Logical and physical extraction
ADB	Android	Android Debug Bridge, basic access
Elcomsoft Mobile Forensics	iOS	For bypassing passcodes and iCloud sync
iTunes Backup	iOS	Can be parsed for data when device is inaccessible

☒ Types of Extraction:

- **Logical:** Extract visible data (contacts, SMS, app data)
- **File System:** Extract entire file system including metadata
- **Physical:** Bit-by-bit clone of the storage (requires root/jailbreak or chip-off)

☒ Mobile Forensics Best Practices:

- Isolate the device (Airplane mode / Faraday bag)
 - Avoid automatic OS updates
 - Disable fingerprint/unlock if legally possible
 - Document everything
 - Extract data using proper tools
 - Always hash backups and images
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☒ Summary Table

Topic	Focus	Key Tools
Live Forensics	Volatile memory, active sessions	Volatility, FTK Imager, Sysinternals
Linux Forensics	Logs, processes, file systems	Autopsy, LiME, Sleuth Kit, bash logs
Mobile Forensics	Apps, call logs, cloud syncs	Cellebrite, Oxygen, Magnet AXIOM